

Lecture I

CMPE240 Digital Systems

Prof. Alper Şen

Today's Class

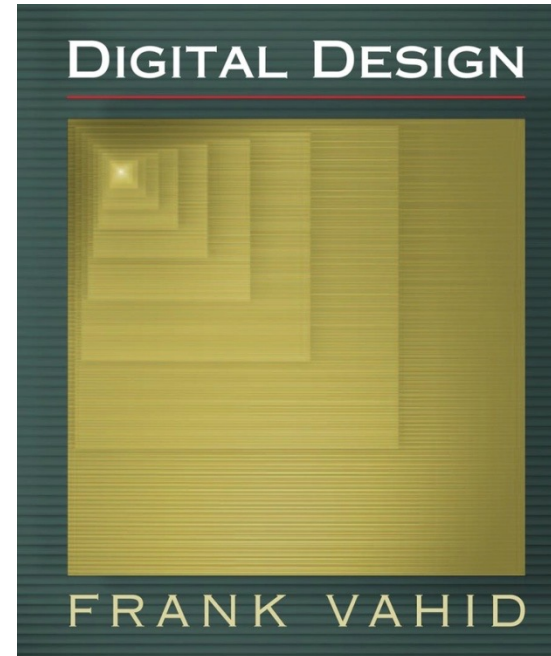
- Course Syllabus
- Course Objectives
- Introduction to Computer Systems

CMPE 240: Digital Systems

- TThTh 512
- **Instructor:** Prof. Alper Şen **Email:** alper.sen@bogazici.edu.tr
(technical/assignment questions must be directed to the class discussion group or TAs)
- **Office Hours:** After class, ETA 25
- **TA:** Emre Bilgili **Email:** ember.bilgili@boun.edu.tr
- **Office Hours:** TBA
- **TA:** **Email:**
- **Office Hours:** TBA
- **TA:** **Email:**
- **Office Hours:** TBA

Course Syllabus

- **Required Textbook:** Frank Vahid, Digital Design, Wiley, 2011,
- Textbook and Student companion site
<http://www.cs.ucr.edu/~vahid/dd/>
- Frank Vahid,
Verilog for Digital Design, Wiley, 2011.
- **Required Lab Book:** CMPE 240 Lab Handbook (soon available on Moodle)



Course Syllabus

- **Supplemental Textbooks (Optional):**
 - John F. Wakerly, Digital Design, Fourth Edition, Prentice Hall, 2006.
 - M. Morris Mano, Digital Design, Fourth Edition, Pearson, 2007.
- **Course Website: Moodle**
<http://www.cmpe.boun.edu.tr/courses/cmpe240/>

Course Syllabus

- **Course Organization:**
 - instructor lectures, TA lab sessions, TA problem sessions, quizzes, exams.
- **Tentative Grading:**
- Midterm I: 25%, Midterm2: 25%, Final: 25%, Labs: 25%, Total: 100%.
- **Right to take the final exam:** Each midterm grade > 15 , Lab grade average > 60 .
- **Problem Sessions:** T8 (or lab hours).

Course Syllabus

- **Course Policies:**
- **Exams:** If your work or a personal situation forces you to unexpectedly miss exams, you should expect to get a zero on those occasions.
- If you miss an exam because of serious injury, you are expected to provide a statement from a doctor stating that, in his/her opinion, it was impossible for you to attend because of injury.
- A slip showing you visited the BU infirmary (revir) or your personal doctor is not sufficient for this. In other situations, you should contact me beforehand.
- **NO MAKEUPS FOR EXAMS, QUIZES, LABS.**

Course Syllabus

- Exams will be performed in class during regularly scheduled class hours.
- Requests for exam date change must be done at least two weeks before the exam. Otherwise, these requests will not be considered even if you have another exam on the same date.

Course Syllabus

- **Cell Phones:** Please turn your cell phone off before coming to the class! Phones are not allowed in labs ! If you bring phone it will be collected at the entrance and we are not responsible for loss !!
- **Late arrival:** Please arrive before the lecture starts.
- **Academic Honesty:** Cheating will not be tolerated.
- You must not show work to any student in the class.
- You must not search for solutions on the internet, copy, study, analyze or even look at any source code produced by other CMPE 240 students from this year, or from years past.
- We will check for cheating, and any incident will be reported to the department.
- **Sign the required Academic Honesty form before the first lab.**

Course Syllabus

- **Labs:**
- The rules for the usage of laboratory will be announced before the first lab.
- Students will automatically get a zero for the labs that they miss. No exception !! No excuses will be accepted.
- NO MAKEUP labs will be given.
- **Lab schedules are going to be fixed soon.**
- **Make sure that we have your email address if you are not already registered in class.**

Questions?

- Your feedback is essential for the course.
- Feel free to ask questions.
- Attend problem sessions and office hours.

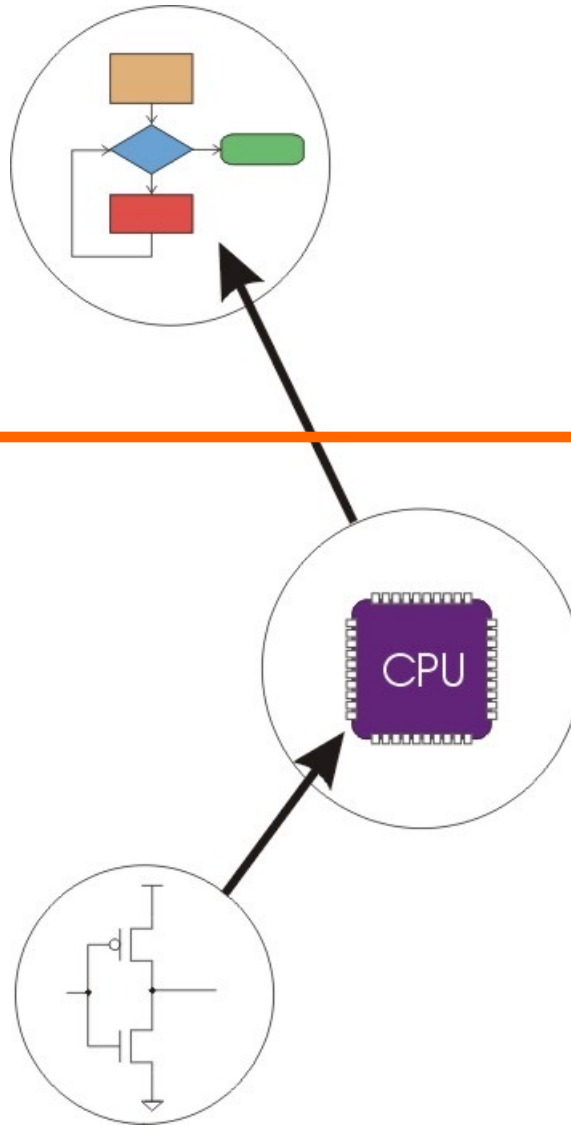
Understanding Computer Systems

- Understanding the whole landscape is crucial
- Gives you a solid understanding, which implies
 - confidence,
 - insight,
 - even makes you a better programmer who is aware of hardware resource issues
- To better understand computer systems, we need to abstract every layer of system development and transformations among layers

Computer System: Layers of Abstraction

Software
CMPE 150
CMPE 160
CMPE 250
CMPE 322

Hardware
CMPE 240
CMPE 344
CMPE443



Problem
Algorithm
Program

Operating System

Instruction Set Architecture

Microarchitecture

Hardware Description

Logic Gates

Circuits

Devices

Transformations Among Layers

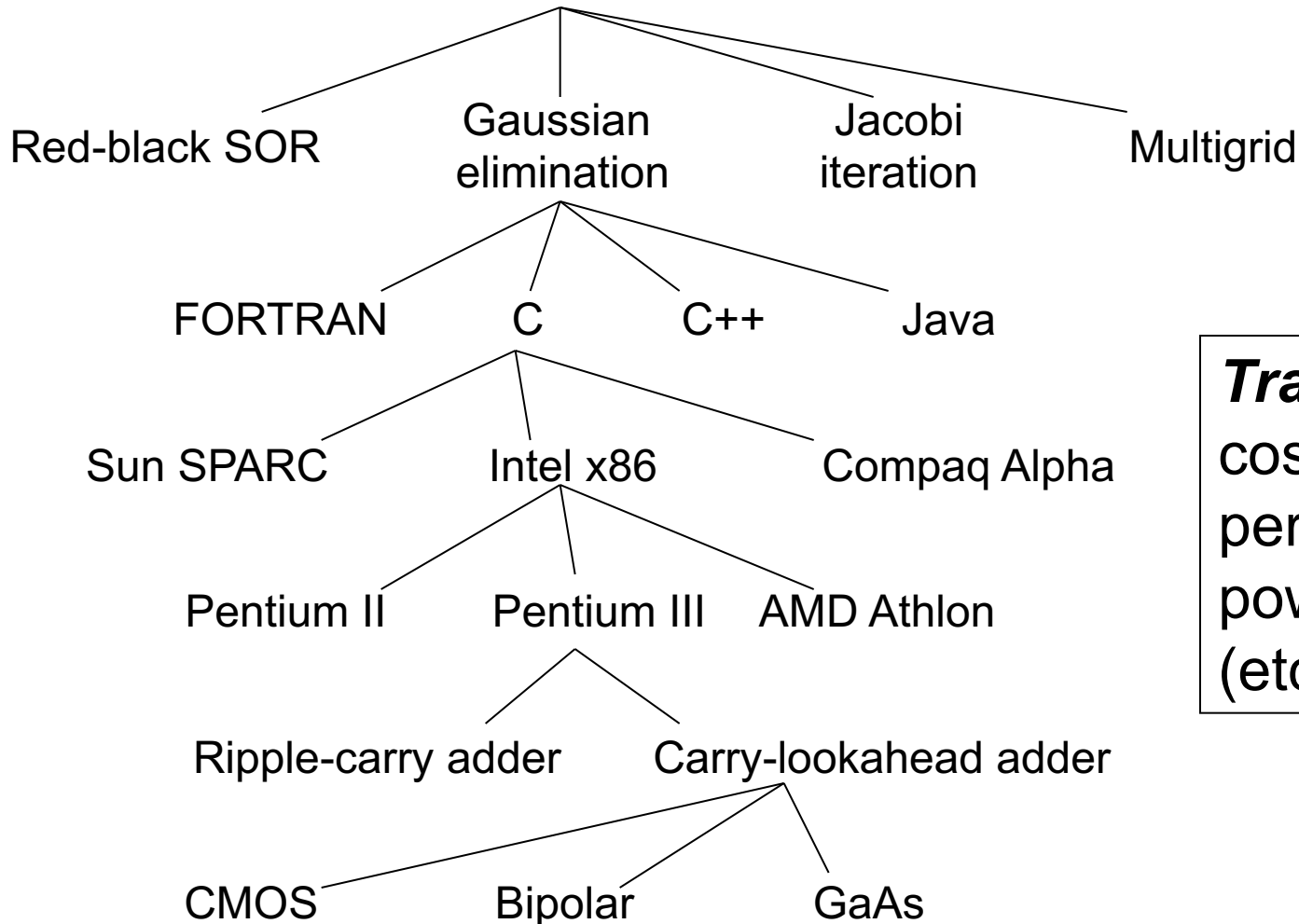
- **Problem Statement:** Add/sort numbers
 - stated using "natural language"
 - may be ambiguous, imprecise
- **Algorithm:** Mergesort, Addition
 - step-by-step procedure, guaranteed to finish, deterministic
 - definiteness, effective computability, finiteness
- **Program:** C, Java, Python
 - express the algorithm using a computer language
 - high-level language, low-level language
- **Operating System:** Linux, Windows, MacOS
 - software that manages hardware resources, provides services for programs (process, file system, networking etc)
- **Instruction Set Architecture (ISA):** x86, ARM, PowerPC
 - specifies the set of instructions the computer can perform
 - data types, addressing mode

Transformations Among Layers

- **Microarchitecture:** Pentium, Core 2, i7, AMD K10
 - detailed organization of a processor implementation
 - different implementations of a single ISA
- **Hardware Description Language:** Verilog, VHDL
 - *description of microarchitecture*
- **Logic Gates/Circuits:** AND/OR/NOT
 - *combine basic operations to realize microarchitecture*
 - *many different ways to implement a single function (e.g., addition)*
- **Devices:**
 - properties of materials, manufacturability (CMOS)

Many Choices at Each Level

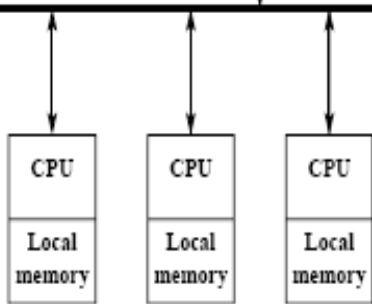
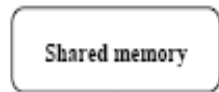
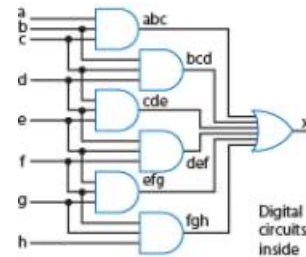
Solve a system of equations



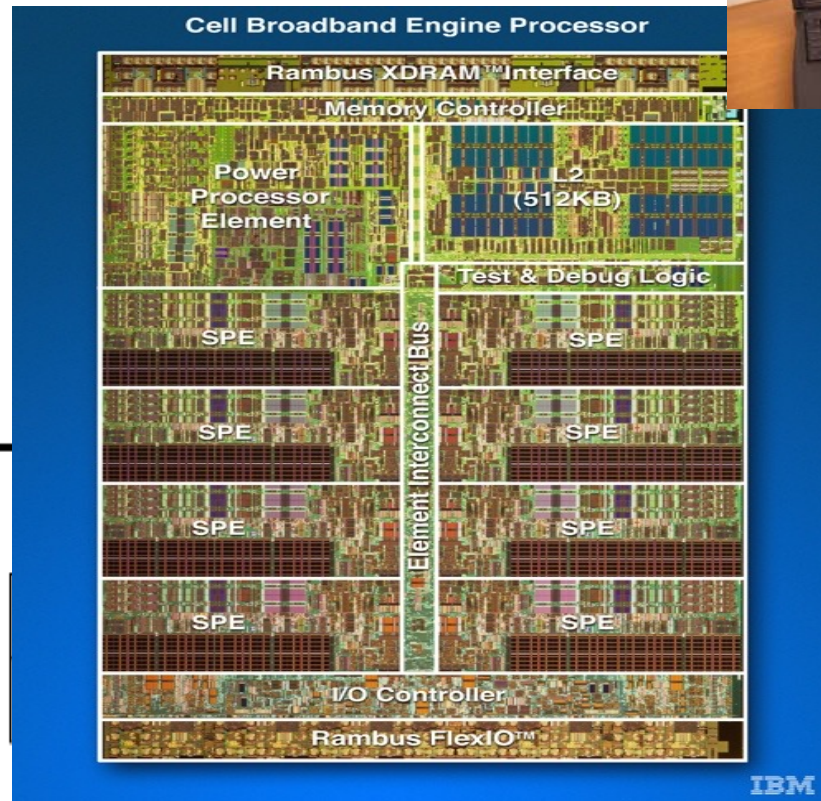
Tradeoffs:
cost
performance
power
(etc.)

Digital Design

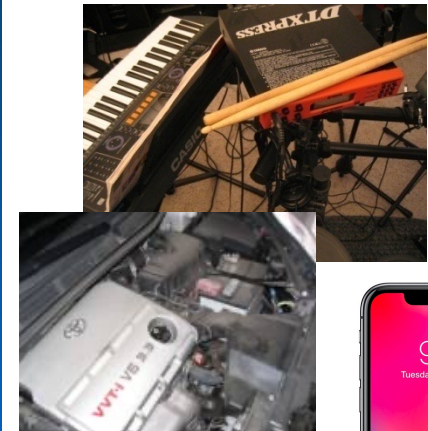
- In this course, learn how to design simple building blocks of computer systems



High Level View



Xbox 360



Course Goals

- Digital systems form the basis of general purpose computers as well as special purpose devices like cell phones, video game consoles, medical instruments, automotive systems, satellites, music equipment, military equipment.
- Goals:
 - introduce the concept of digital and binary systems
 - provide fundamental concepts used in the design and analysis of digital systems
 - give students the basic tools for the design and implementation of digital circuits
 - to be able to model basic digital circuits in hardware description languages (Verilog)
 - reinforce theory and techniques through labs

Course and Lab Schedule and Contents (Tentative)

Order and contents may change.

Topics	Reading (Vahid)
Introduction	Chapter 1
Combinational logic design: Number systems, codes and arithmetic. Boolean algebra and logic gates. Boolean functions, Canonical and Standard Forms. Simplification.	Chapter 2
Sequential logic design: Flip-flops, FSMs,	Chapter 3
Combinational and sequential component design: adders, shifters, comparators, multipliers, subtractors, arithmetic logic units, registers, counters, register files	Chapter 4
Register Transfer Level Design of digital circuits	Chapter 5
Optimizations and Tradeoffs	Chapter 6
Physical Implementation: FPGAs	Chapter 7

Summary

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